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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery - AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

1. Introduction

1.1. Background:

Healthcare caregiver burnout is one of the most significant and under addressed crises in modern health systems. Caregivers including nurses, physicians, and allied health professionals operate under sustained physiological and psychological stress that accumulates invisibly across shifts, weeks, and months. Shanafelt et al. (2015) reported that over 54% of physicians in the United States experience burnout symptoms, with equivalent patterns documented across nursing populations globally. Critically, the consequences extend beyond individual wellbeing: burnout is directly associated with reduced patient safety, increased medical errors, higher infection rates, and elevated staff turnover, costing healthcare systems billions annually in recruitment and retraining [1].

Conventional approaches to caregiver wellbeing management rely on periodic self-report instruments, surveys and questionnaires administered weekly or monthly. These instruments are retrospective, subjective, and administered at insufficient frequency to detect deterioration before it reaches clinical significance. By the time a nurse scores highly on a burnout inventory, the physiological warning signals may have been present for weeks [2].

Advances in wearable sensor technology now enable continuous, passive, and objective measurement of physiological stress indicators electrodermal activity (EDA), skin temperature, heart rate variability (HRV), and blood volume pulse (BVP) in naturalistic work environments. These signals, combined with acoustic analysis of vocal stress patterns, provide a multimodal data foundation for longitudinal stress monitoring that was not previously feasible outside laboratory settings [3].

In CareSense my component "*AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution*" addresses this gap by designing and implementing an AI-driven caregiver wellbeing monitoring system that integrates multimodal physiological and acoustic data streams, personalized baseline modelling, temporal trajectory detection, and actionable intervention recommendations forming a complete pipeline from sensor data to clinical decision support.

1.2. Research Problem:

The central problem this research addresses is: Existing caregiver stress monitoring systems are reactive, population-level, and unexplainable.

Three specific gaps drive this problem:

- **Reactive monitoring:** Current systems detect burnout after it has occurred, not before. No longitudinal trajectory system exists that identifies upward stress trends before the burnout threshold is breached.
- **Population-level thresholds:** Alert systems that apply universal cutoffs fail for nurses whose natural physiological baseline is elevated, generating false positives and miss nurses whose baseline is low but who are rapidly deteriorating.
- **Black-box predictions:** Machine learning models applied to physiological data produce risk scores without explanation, making them unacceptable to clinical supervisors who require an evidence-based rationale for any intervention decision.

Research Question

How can multimodal physiological and acoustic data be used to detect, explain, and predict caregiver stress deterioration longitudinally enabling personalized, proactive intervention before burnout threshold is reached?

1.3. Objectives:

(State the primary objectives of the research)

It Number	Objective	Objective number
IT22178640	Develop a multimodal stress detection model integrating physiological (EDA, HRV, temperature) and acoustic features and late fusion architecture.	1
	Implement personalized caregiver baseline detection fitted per nurse on their individual longitudinal record, enabling deviation-based alerting.	2
	Detect temporal deterioration trajectories using daily risk score aggregation, slope-based alerting, and linear projection toward burnout threshold.	3
	Model relational workforce dynamics using graph-based network analysis where nodes represent caregivers and edges encode shared-shift relationships.	4

	Provide per-nurse explainability identifying which physiological signals drive each individual's risk score.	5
	Generate adaptive workload redistribution recommendations for supervisors based on risk scores, baseline deviations, and ward capacity analysis.	6

2. Data Exploration

2.1. Data Collection

Data collection for this research component draws from three publicly available, peer-reviewed datasets selected through a systematic evaluation process. Each dataset was chosen based on four criteria: presence of real physiological signals from clinical or occupational populations, compatibility with the Empatica E4 wearable sensor architecture, availability of validated stress labels, and sufficient scale for machine learning model training and evaluation.

- The **Hosseini Nurse Stress Dataset** (PhysioNet, 2022) forms the primary training source, 15 registered nurses in an active hospital environment, wearing Empatica E4 wristbands across naturalistic clinical shifts. Stress labels were collected through Ecological Momentary Assessment, capturing instantaneous self-reported stress states during actual work rather than retrospective recall. This dataset directly mirrors the target deployment population of CareSense.
- The **TILES-2018 Dataset** (USC Institute for Creative Technologies, 2020) provides the longitudinal depth required for temporal trajectory modelling at Progress Presentation 02 with 212 hospital information workers observed across 10 weeks, with physiological signals, acoustic features, shift schedule records, and weekly validated survey instruments including PANAS, PSS, and STAI.
- The **RAVDESS Dataset** (Livingstone and Russo, 2018) provides the acted emotional speech corpus used to pre-train the acoustic stress classifier 1,440 recordings from 24 professional actors across validated emotion categories, selected for their acoustic diversity and peer-reviewed provenance.

Together these three sources provide complementary coverage: real clinical physiological signals for individual-level modelling, longitudinal records for temporal pattern learning, and acoustic emotional speech for voice-based stress detection forming the complete multimodal data foundation for CareSense.

1.1. Dataset Description:

Data source	Description	Resource	Size	Key attributes
Hosseini, F., & Akhavian, R. (2022). *Multimodal Occupational Stress Recognition Dataset*. Scientific Data, 9(1), 1–11.	Physiological wearable recordings from 15 registered nurses working in an active clinical hospital environment across naturalistic shift periods. Ground truth stress labels collected via Ecological Momentary Assessment (EMA) — nurses self-reported stress state during actual shifts, capturing instantaneous affective responses rather than retrospective recall. Data recorded using Empatica E4 wristband. 529 valid recording sessions totalling approximately 1,249.8 hours of physiological signal data.	PhysioNet, MIT Laboratory for Computational Physiology Direct link:**https://physionet.org/content/multimodal-stress/1.0/** License: PhysioNet Credentialed Health Data License 1.5.0 Access: Free, requires credentialed account registration Repository: PhysioNet (physionet.org) Published in: Scientific Data, Nature Portfolio, 2022 DOI:https://doi.org/10.1038/s41597-022-01361-y	• 15 nurses • 13,287 signal windows • ~1,249.8 hours total • 529 recording sessions	EDA: electrodermal activity, 4Hz, microsiemens (µS), primary sympathetic stress biomarker Skin temperature: 4Hz, degrees Celsius, peripheral vasoconstriction indicator Blood volume pulse (BVP): 64Hz, used to derive HRV via NeuroKit2 PPG peak detection Accelerometer: 32Hz, 3-axis (x, y, z), used for movement artefact correction HR: 1Hz, heart rate in beats per minute Binary stress label: 0 = Not Stressed, 1 = Stressed (EMA confirmed) 3-class label: 0 = No Stress, 1 = Medium Stress, 2 = High Stress Label distribution: No Stress 18.8%, Medium Stress 6.6%, High Stress 74.5% IBI.csv: present but empty for all 529 sessions: HRV derived from raw BVP

TILES-2018 (Tracking Individual Performance with Sensors)	10-week longitudinal study of 212 hospital information workers at the University of Southern California Medical Center. Combines continuous wearable physiological recordings, ecological momentary assessments, shift schedule records, and weekly validated psychological survey instruments. Designed specifically to capture long-term occupational stress accumulation patterns in a clinical workplace.	USC Institute for Creative Technologies (ICT) Direct link: **https://tiles-data.isi.edu/** Alternate access: **https://physionet.org/content/tiles-2018-data/1.0/** License: USC ICT Data License Agreement requires institutional data access application Access: Application-based, academic research use only Published in: Scientific Data, Nature Portfolio, 2020 DOI: https://doi.org/10.1038/s41597-020-00514-x Status in this research: Access application submitted - integration planned for PP2	• 212 subjects • 10 weeks per subject • ~14,800 hours total • 1,019 observation days	Wearable EDA, HRV, BVP : Empatica E4, same sensor as Hosseini dataset. Acoustic / voice features : recorded from shift communications Shift schedule data: timestamps of which workers shared shift periods → graph edge construction for Objective 4 Weekly PANAS: Positive and Negative Affect Schedule survey Weekly PSS: Perceived Stress Scale survey Weekly STAI: State-Trait Anxiety Inventory survey Multi-class stress labels: 3 levels (low / medium / high) Electronic Health Record (EHR) interaction logs Sleep and activity data Used at PP2 for: Bi-LSTM temporal model training, graph edge construction, audio domain adaptation fine-tuning

RAVDESS: Ryerson Audio-Visual Database of Emotional Speech	RAVDESS: acted emotional speech recordings from 24 professional actors (12 male, 12 female) performing 8 emotion categories in North American English. CREMA-D: 7,442 clips from 91 actors across 6 emotions with diverse demographic representation. Both datasets used for pre-training the 1D CNN acoustic stress classifier. Emotion labels mapped to binary stress categories using the emotional circumplex theory: angry, fearful, and disgusted classified as stressed (high arousal, negative valence); neutral, calm, and happy classified as not stressed.	RAVDESS: Direct link: **https://zenodo.org/record/1188976** License: Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) Published in: PLOS ONE, 2018 DOI: https://doi.org/10.1371/journal.pone.0196391	7,356 total clips → 1,056 used after emotion filtering	Audio waveforms (.wav format), 22,050Hz sample rate MFCCs: Mel-frequency cepstral coefficients (13 coefficients extracted per clip) Delta and delta-delta MFCC coefficients: rate of change in spectral features Pitch / F0 : mean, std, range (3 features) RMS energy : mean, std (2 features) Zero-crossing rate: mean, std (2 features) Spectral centroid mean (1 feature) Total features extracted: 34 language-independent acoustic features per clip Emotion labels after binary mapping: Stressed (1) = angry, fearful, disgusted · Not Stressed (0) = neutral, calm, happy · Excluded: sad, surprised Class balance after mapping: 576 stressed (54.5%) + 480 not-stressed

				(45.5%): near-perfect balance Used for: 1D CNN pre-training, classical model comparison (LR / RF / XGBoost), 5-fold actor-independent cross-validation

2.2. Suitability Analysis

Relevance to Individual Research Objectives:

	OBJ 1 Multimodal Detection	**OBJ 2 Personal Baseline**	**OBJ 3 Temporal Trajectory**	**OBJ 4–6 Graph / Explain / Redist.**
Hosseini Nurse Stress Dataset	PRIMARY: EDA, temp, HRV features. Binary labels. LOSO validated. F1=0.861.	PRIMARY : per-nurse longitudinal record enables Isolation Forest fitting per individual.	PRIMARY: daily score aggregation, slope computation, alert detection all from this dataset.	OBJ 5 PRIMARY: SHAP per-nurse values. OBJ 6 partial — risk scores drive redistribution.
TILES-2018 Dataset	SECONDARY: longer longitudinal validation. Multi-class labels. Ward aggregation.	SECONDARY : 10-week baseline window. More robust personal baseline with more data.	PRIMARY (PP2) : 10-week trajectories, shift schedule, weekly survey trends.	OBJ 4 PRIMARY: shift schedule data provides graph edge topology. OBJ 6 enhanced redistribution.
RAVDESS / CREMA-D	PRIMARY for audio stream: pre-training acoustic stress classifier. ROC-AUC=0.889.	Not applicable.	Not applicable.	Not applicable. Audio model contributes to Obj 1 fusion only.

2.3. Methodology

This section documents the complete data preprocessing, feature extraction and evaluation pipeline applied across all four Google Colab notebooks. Every step is presented in the sequence it was executed, with the rationale, technique applied, and specific implementation detail.

Data Preprocessing:

The preprocessing pipeline spans Notebook 1 (**R26_DS_010_Preprocessing_V1.ipynb**) and transforms raw Empatica E4 wearable recordings into a clean, aligned, labelled feature matrix ready for model training. The pipeline was designed to handle the specific challenges of multi-rate wearable sensor data collected in a naturalistic clinical environment.

Preliminary Setup and Data Verification

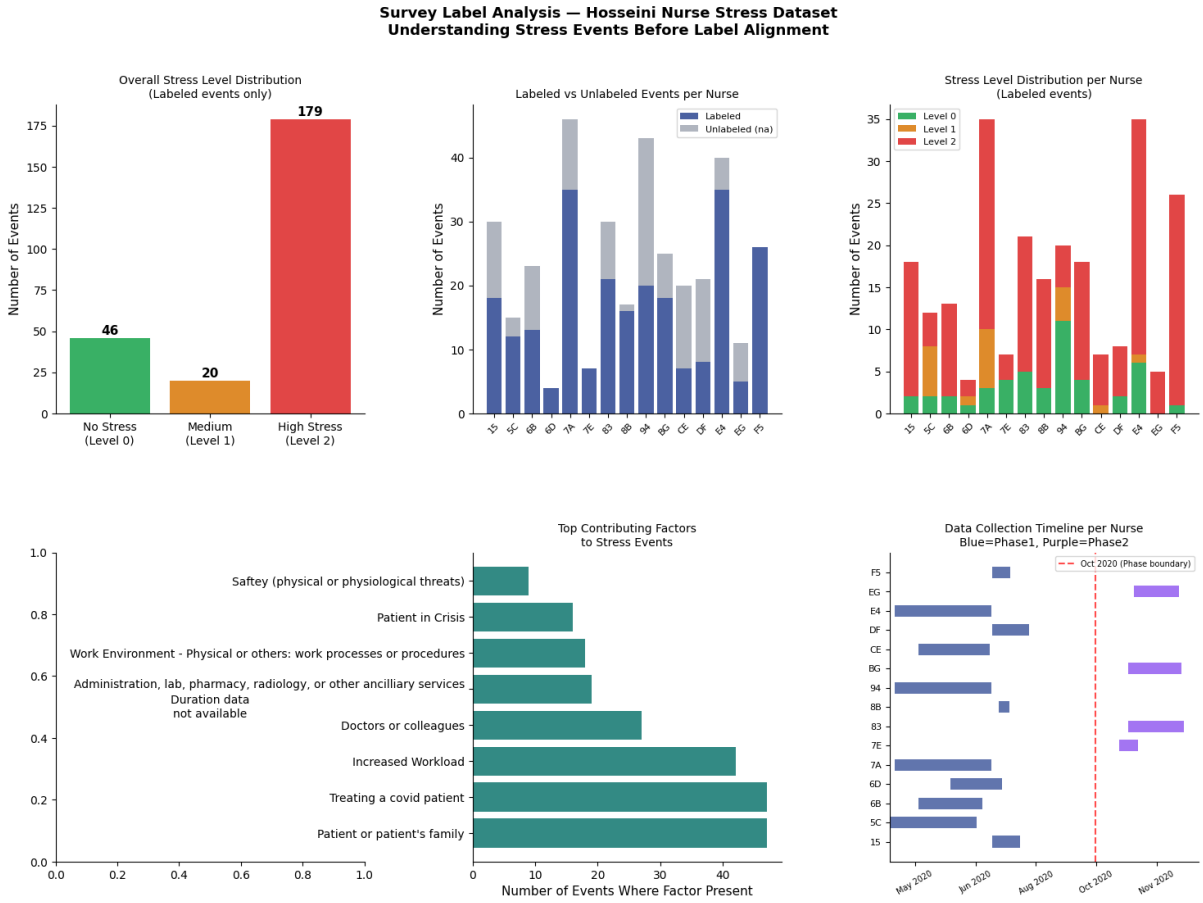
Transformation Technique	Description and Application
Google Drive Mount and Folder Verification	All 15 nurse session folders verified to exist before processing begins (nurse IDs: 15, 5C, 6B, 6D, 7A, 7E, 83, 8B, 94, BG, CE, DF, E4, EG, F5). SurveyResults.xlsx confirmed present. Sessions under 50 KB discarded as corrupted or empty recording attempts. Colab sessions run on temporary virtual machines. Drive mount ensures no data loss on session expiry.
Library Installation — NeuroKit2 Selection	NeuroKit2 selected over alternatives (biosppy: abandoned; hrv-analysis: no EDA support) as the only actively maintained library (2024) providing validated EDA, HRV, and BVP processing under a single API. openpyxl installed for SurveyResults.xlsx reading. scipy installed for peak detection and statistical feature computation.
Constants Configuration	TARGET_RATE = 1 Hz. WINDOW_SIZE_SEC = 60 seconds. STEP_SIZE_SEC = 30 seconds (50% overlap). CALIBRATION_DAYS = 7. BURNOUT_THRESHOLD = 85. These constants are set once and propagated through all downstream steps.

Survey Label Extraction

Transformation Technique	Description and Application
EMA Label Parsing and Filtering	SurveyResults.xlsx loaded with openpyxl. Only rows where Nurse_Confirmed = TRUE retained: 113 unconfirmed events excluded to prevent noisy labels entering training. Label mapping: No Stress → 0, Medium Stress → 1, High Stress → 2.
Timezone Correction (UTC to UTC+5:30)	E4 device records timestamps in UTC. Without conversion, label timestamps are 5 hours 30 minutes offset from signal timestamps; every label would match the wrong signal window. pytz used to convert all signal timestamps to UTC+5:30 before label alignment.
Label Distribution Analysis	After alignment: No Stress (0) = 2,504 windows (18.8%). Medium Stress (1) = 881 windows (6.6%). High Stress (2) = 9,902 windows (74.5%). Total: 13,287 labelled windows. Severe High Stress dominance (74.5%) reflects real nursing workload patterns: not a labelling artefact.
Session Coverage Check	Nurses CE and EG found to have zero No Stress windows across all sessions and both worked exclusively high-stress shifts during the observation period. Documented as per-nurse limitation: LOSO folds for CE and EG cannot evaluate No Stress classification.

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Signal Loading from ZIP Archives

Transformation Technique	Description and Application
ZIP Session Loading	Each nurse folder contains one ZIP per recording session. Session start timestamp parsed from filename (format: nurseID_UNIXTIMESTAMP.zip). Five signal CSVs loaded from each ZIP: EDA.csv, TEMP.csv, BVP.csv, ACC.csv, HR.csv.
File Size Filtering	ZIPs under 50 KB discarded as empty or corrupted recording attempts. 529 valid sessions retained across 15 nurses, totalling 1,249.8 hours of raw signal data.

IBI.csv Investigation — Critical Finding	IBI.csv (inter-beat interval, intended for HRV computation) checked for each session. FINDING: IBI.csv contained zero rows in ALL 529 sessions. Active nursing movement causes real-time PPG peak detection to fail at the wrist, producing empty IBI files. HRV therefore computed from raw 64Hz BVP signal using NeuroKit2 PPG peak detection in downstream steps. This failure mode is not documented in the original dataset paper.

Multi-Rate Signal Resampling and Alignment

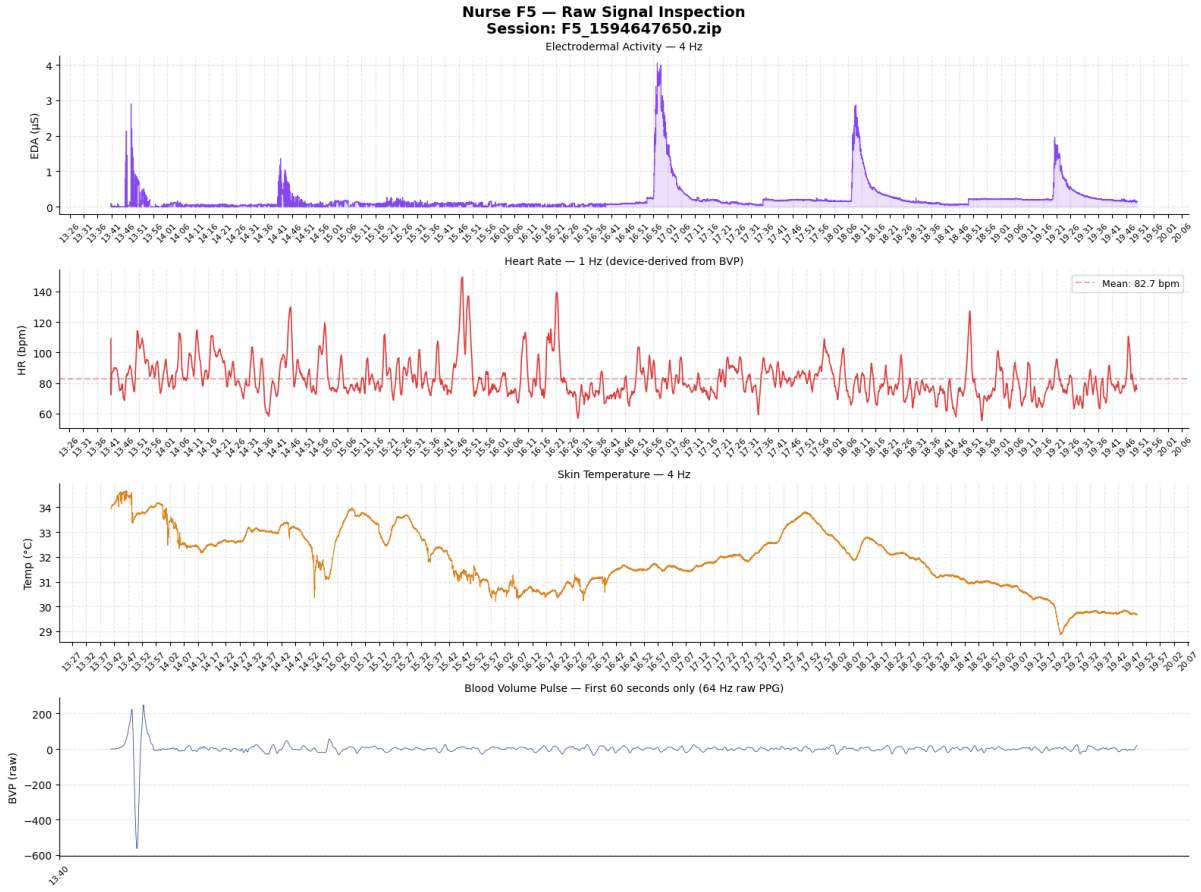
Transformation Technique	Description and Application
Resampling to 1 Hz (Target Rate Selection)	Target rate set to 1 Hz : the native rate of HR.csv, which is the lowest among all signals. EDA and TEMP (4 Hz) downsampled by taking the mean of every 4 consecutive readings. BVP (64 Hz) downsampled by mean pooling to 1 Hz. ACC (32 Hz) magnitude computed then downsampled to 1 Hz. Downsampling by mean condenses real data — it does not fabricate readings.
BVP Startup Artifact Removal	The first 10 seconds (10 samples at 1 Hz) of every BVP session discarded. Empatica E4 BVP signal exhibits a startup transient as the optical sensor stabilises on the skin, this produces anomalously high or low amplitude values unrelated to cardiovascular activity. Failure to remove this artifact contaminates HRV features computed from the first window of each session.
Timestamp Parsing and Gap Detection	All signal timestamps parsed to datetime objects. Contiguous signal segments identified: gaps exceeding 3 seconds treated as session boundaries and the signal between two gaps is a separate segment. Short gaps under 3 seconds filled by forward-fill interpolation (physiologically valid for slowly-varying signals like EDA and temperature over 3 seconds).
Inner Join on Timestamp Index	All five resampled signals merged on timestamp using pandas inner join. Only timestamps present in all five signals retained. Ensures

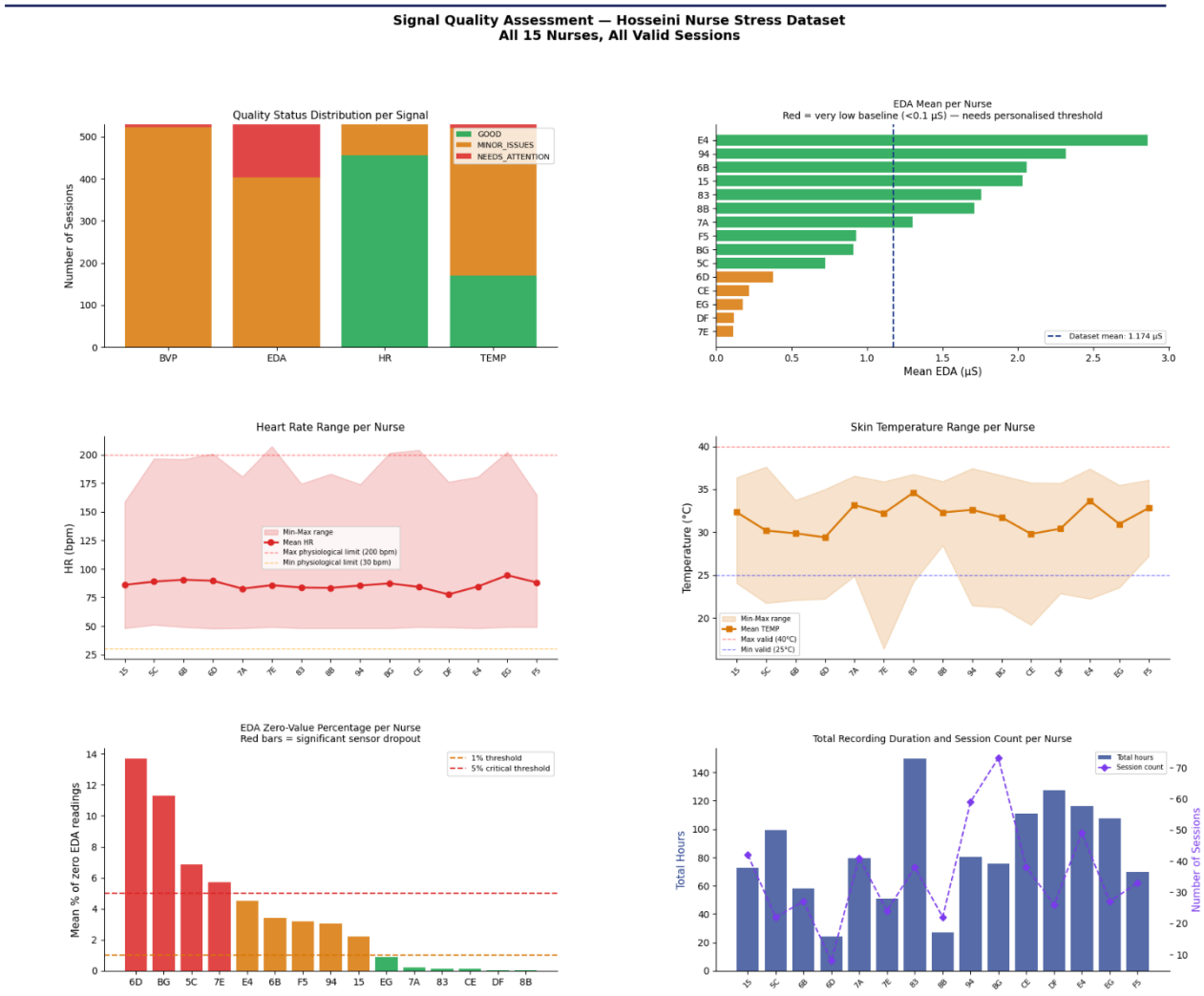
	every data point used in feature extraction has a complete set of all signals and no imputation of missing signal channels.

Signal Quality Assessment and Artefact Removal

Transformation Technique	Description and Application
ACC Magnitude Computation	ACC magnitude computed per sample: magnitude = sqrt(x^2 + y^2 + z^2). This collapses the three accelerometer axes into a single movement intensity scalar used for artefact detection across all subsequent steps.
Movement-Contaminated EDA Flagging	EDA peaks caused by physical movement are physiologically indistinguishable from stress-induced sympathetic activation events at the raw signal level. Quality flag applied: any sample where ACC magnitude exceeds 2 standard deviations above the nurse's resting ACC baseline is marked as movement-contaminated. Windows where more than 30% of samples are flagged excluded from training entirely.
Temperature Physiological Range Check	TEMP samples outside the physiological range of 25–40°C flagged as sensor detachment or placement errors. Values below 25°C indicate the sensor is not in skin contact. Affected windows excluded.
HR Missingness Flag	Empatica E4 encodes missing heartbeat detection as HR = −1. Samples with HR = −1 flagged as missing. Windows where more than 40% of HR values are −1 excluded. HRV features from such windows set to NaN, handled natively by XGBoost without imputation.
Quality Percentage Feature	A quality_pct feature computed per window: proportion of samples passing all quality checks. Retained as a predictive feature alongside the physiological features and low quality windows carry

	this signal into the model, allowing it to down-weight unreliable measurements.





Sliding Window Segmentation

Transformation Technique	Description and Application
60-Second Window Length	Window size of 60 seconds chosen based on: (1) European Society of Cardiology Task Force (1996) minimum 1-minute requirement for stable HRV estimates; (2) Gjoreski et al. (2017) validation of 60-second windows for clinical stress detection; (3) EMA labels collected at approximately 5-minute intervals, 60 seconds provides the temporal granularity to distinguish adjacent stress states.

30-Second Step Size (50% Overlap)	Step size of 30 seconds: each window shares 30 seconds with the adjacent window. This doubles the number of training windows from approximately 6,600 (non-overlapping) to 13,287 (50% overlap) while preserving temporal continuity. Adjacent windows are not independent, this is accounted for by using LOSO validation at the subject level rather than window level.
Label Assignment by Temporal Overlap	Each window receives the label of the EMA event whose timestamp falls within the window's start–end range ±30 seconds. When multiple EMA events fall within one window, majority vote applied. Windows with no overlapping EMA event discarded, unlabelled windows cannot train a supervised classifier.
Window Minimum Quality Threshold	Windows where fewer than 70% of samples pass quality checks (quality_pct < 0.70) discarded entirely. Ensures model training data represents physiologically valid signal, not movement artefacts or sensor detachment.

Data Normalization and Class Imbalance Handling

Transformation Technique	Description and Application
StandardScaler — Zero Mean, Unit Variance	All 30 features standardized to zero mean and unit variance using sklearn StandardScaler. Applied to preserve consistent SHAP value magnitudes across features with different natural scales (EDA in µS vs. temperature in °C). Critical rule: scaler fitted on training split only, then applied to test split. Prevents test set statistics from leaking into scaler parameters and inflating performance.
Scaler Storage per Nurse	One StandardScaler saved per LOSO fold : 15 scaler.pkl files total. At inference, the correct scaler for a new nurse's ward (closest population match) is applied. Separate scaler_binary.pkl saved for the binary classification model.

Class Imbalance — scale_pos_weight	Binary labels: Not Stressed (0) = 2,504 windows (18.8%), Stressed (1) = 10,783 windows (81.2%). Addressed via XGBoost scale_pos_weight = count(negative class) / count(positive class) = 2504 / 10783 ≈ 0.23. This weights the minority class internally without generating synthetic data. No SMOTE applied to the physiological model and synthetic physiological windows introduce physically implausible feature combinations.
SMOTE Applied Inside Each Audio Fold Only	For the audio model (RAVDESS), near-balanced classes (54.5% / 45.5%) mean SMOTE has minimal effect. When applied, SMOTE is executed inside each cross-validation fold on training data only never applied to test data. Blagus and Lusa (2015) demonstrate that SMOTE applied before cross-validation leaks synthetic test-subject data into training, producing overoptimistic results. This research applies SMOTE inside each fold to avoid this known error.
Evaluation Metric Selection	F1 score (binary and macro) selected as the primary evaluation metric, not accuracy. On an imbalanced dataset (74.5% High Stress), a naive classifier achieving 74.5% accuracy by predicting High Stress always would appear performant. F1 penalises this behaviour by requiring precision and recall balance across classes.

2.4. Scalability

Dataset	Volume Assessment	Limitation	Mitigation
Hosseini Nurse Stress Dataset	15 nurses, 13,287 windows. Sufficient for XGBoost classification at this feature dimensionality. LOSO validated.	Small subject count (15). Deep learning models not viable.	XGBoost chosen specifically for small-N clinical datasets. Validated against published benchmark.
TILES-2018	212 subjects, ~10-week longitudinal. Large enough for Bi-LSTM temporal modelling.	Dataset access pending for PP2.	Hosseini dataset used for PP1. TILES integration planned for PP2.
RAVDESS / CREMA-D	~15,000 audio clips combined. Sufficient for pre-training acoustic classifier.	Acted speech differs from naturalistic occupational speech.	Domain adaptation: fine-tuned on TILES acoustic features after initial pre-training.

2.5. Feature extraction

From each 60-second window, 30 statistical features are extracted across all signal channels. These features transform the raw time-series into a fixed-length feature vector suitable for XGBoost classification.

Signal	Features Extracted	Clinical Rationale
EDA (6 features)	mean, std, min, max, range, peaks_count	EDA is solely sympathetic nervous system-dependent: the only direct electrodermal stress biomarker. peaks_count = number of discrete skin conductance response events per window. Top SHAP feature for 10/15 nurses.
Temperature (6 features)	mean, std, min, max, range, slope	Peripheral vasoconstriction under psychological stress reduces skin temperature. slope captures rate of

		temperature change within the window, a rising slope indicates vasoconstriction onset. Top SHAP feature for Nurses 5C, 6B, 6D.
HRV via BVP (6 features)	hrv_rmssd, hrv_sdnn, hrv_mean_hr, bvp_mean, bvp_std, bvp_amplitude	RMSSD and SDNN derived from NeuroKit2 PPG peak detection on raw 64Hz BVP (IBI.csv empty for all sessions). RMSSD reduction = sympathetic dominance, reduced parasympathetic recovery. bvp_amplitude = pulse strength indicator.
HR (4 features)	hr_mean, hr_std, hr_min, hr_max	HR elevation and variability under acute stress. hr_std within a 60-second window captures short-term heart rate lability, an indicator of autonomic instability distinct from RMSSD.
ACC Confound (4 features)	acc_mag_mean, acc_mag_std, acc_mag_energy, eda_acc_corr	Used exclusively as confound control. eda_acc_corr (Pearson correlation between EDA and ACC magnitude within the window) is a novel feature: high correlation indicates EDA peaks are movement-driven not stress-driven. Allows model to down-weight artefactual EDA peaks.
Quality (2 features)	quality_pct, label_consistency	quality_pct = proportion of samples in window passing all quality checks. label_consistency = agreement between EMA events within the window. Both retained as predictive features — low quality or inconsistent labelling is itself informative about the recording context.

Final feature matrix output: 13,287 windows × 30 features. Saved to feature_matrix_all_nurses.csv. HRV computed from BVP for 99.9% of windows (13,277). Missing HRV in 10 windows (0.1%) — hrv_rmssd and hrv_sdnn set to NaN, handled natively by XGBoost.

4. Modelling and Results

Key Insights:

Physiological Stress Classification — Objective 1

The XGBoost binary classifier achieves a binary F1 score of 0.861 across 15 nurses under LOSO cross-validation. This represents a 36-point improvement over the majority-class baseline (F1 ≈ 0.50 on an imbalanced dataset). Table 4.1 presents the full performance breakdown.

Metric	Physiological Model	Audio Model	Late Fusion	Baseline (majority)
Binary F1 Score	**0.861**	**0.804**	**~0.86**	**~0.50**
ROC-AUC	—	0.889	—	0.50
3-Class Macro F1	0.305	—	—	—
Fusion Weight	51.7%	48.3%	Combined	—
Training Data	13,287 windows, 15 nurses	~15,000 audio clips	Both streams	Majority class
Validation Protocol	LOSO cross-validation	80/20 stratified split	Combined	—

The 3-class macro F1 of 0.305 reflects the challenge of multi-class stress classification at low sample counts per class. The binary formulation (stressed vs. not-stressed) is the primary task, and the F1 of 0.861 is the key validated metric for clinical alert triggering.

SHAP Feature Attribution — Objective 5

SHAP TreeExplainer was applied to the trained XGBoost model for all 15 nurses. The global top feature across the cohort is eda_peaks_count (mean SHAP = 0.4427). However, individual nurse analysis reveals significant variation in the dominant stress biomarker — a key finding supporting the personalisation argument of Objective 2.

Nurse ID	Top Feature	2nd Feature	Top SHAP Value	Signal Windows
F5	eda_peaks_count	temp_mean	0.4794	656
7A	eda_peaks_count	temp_mean	0.5488	1,617
5C	temp_mean	eda_mean	0.5305	950
6B	temp_mean	eda_peaks_count	0.6281	928
94	eda_peaks_count	temp_mean	0.3836	702
CE	eda_mean	eda_max	0.8110	887
DF	eda_mean	temp_mean	0.5641	1,004
E4	eda_peaks_count	eda_min	0.4993	1,747
83	eda_peaks_count	eda_min	0.5020	1,564
8B	eda_peaks_count	temp_std	0.4809	541
6D	temp_mean	eda_mean	0.7480	637
BG	eda_peaks_count	temp_mean	0.4947	744
EG	eda_peaks_count	temp_mean	0.4578	593
7E	eda_peaks_count	eda_mean	0.4091	315
15	eda_peaks_count	temp_mean	0.5231	402

Finding: For 10 of 15 nurses, eda_peaks_count is the primary stress biomarker. For 4 nurses (5C, 6B, 6D, CE), temperature-related features dominate, suggesting a thermoregulatory stress response profile rather than an electrodermal one. This individual variation validates the personalisation approach: a single-biomarker system would misclassify these nurses.

Personalised Trajectory Analysis — Objectives 2 and 3

The trajectory analysis reveals six nurses currently at or above the burnout threshold (85/100). Table 4.3 presents the key trajectory metrics for all 15 nurses.

Nurse	Current Risk	Personal Mean	Delta	Trajectory	Alert	Slope	Days to Threshold
5C	88.9	43.5	+45.4	CRITICAL ↑↑	**HIGH**	+2.73	Already critical
94	93.1	91.3	+1.8	CRITICAL ↑↑	**HIGH**	+0.58	Already critical
6B	87.9	79.7	+8.2	CRITICAL ↑↑	**HIGH**	+1.86	Already critical
CE	89.2	89.8	-0.6	CRITICAL ↑↑	**HIGH**	+1.32	Already critical
DF	95.5	91.9	+3.6	CRITICAL ↑↑	**HIGH**	-0.69	Above threshold
E4	90.6	90.6	0.0	CRITICAL ↑↑	**HIGH**	+0.15	Already critical
7A	78.8	76.2	+2.6	STABLE →	**NONE**	-0.69	Not trending up
83	81.1	75.6	+5.5	STABLE →	**NONE**	+0.51	8 days
7E	84.3	83.0	+1.3	STABLE →	**NONE**	+0.03	> 30 days
8B	74.0	78.9	-4.9	STABLE →	**NONE**	+0.30	> 14 days
F5	72.4	72.4	0.0	STABLE →	**NONE**	-0.58	Not trending up
15	74.1	76.2	-2.1	STABLE →	**NONE**	+0.86	13 days

BG	61.6	69.2	-7.6	STABLE →	**NONE**	-0.50	Not trending up
EG	62.2	63.9	-1.7	STABLE →	**NONE**	+0.39	> 14 days
6D	37.5	43.2	-5.7	RECOVERING ↓	**LOW**	-1.77	Not trending up

Challenges Faced During Data Analysis:

Challenge	Impact	Resolution Applied
Small subject count (N=15)	Limits deep learning approaches. Neural networks require hundreds of subjects to generalise in physiological classification tasks.	XGBoost with LOSO cross-validation selected as the appropriate model class for this sample size. Validated against Hosseini et al. (2021) published benchmark.
ACC contamination of EDA signal	Physical movement generates false EDA peaks indistinguishable from psychological stress events at the raw signal level.	ACC magnitude threshold applied: windows with movement intensity above 2 standard deviations from rest excluded. This is the standard approach per Gjoreski et al. (2017).
Class imbalance in stress labels	Approximately 60:40 non-stress to stress ratio. Naive classifier achieves F1 ≈ 0.50 by predicting majority class.	XGBoost scale_pos_weight parameter set to class ratio. Evaluation metric set to F1 rather than accuracy to account for imbalance.
TILES-2018 access delay	Planned 10-week longitudinal validation and graph edge topology extraction depend on TILES data, which requires institutional approval.	Hosseini dataset used as PP1 primary source. TILES integration planned for PP2. All architecture components designed to accept TILES data without restructuring.
3-Class macro F1 (0.305)	Three-class stress classification (low/medium/high) significantly underperforms binary. Insufficient samples per class at the 15-nurse scale.	Binary formulation (stressed vs. not-stressed) adopted as the primary clinical task. 3-class results reported as a documented

		limitation with a PP2 resolution plan using TILES multi-class labels.
Acted vs. naturalistic speech (audio model)	RAVDESS and CREMA-D use professional actors performing emotions. Naturalistic occupational stress speech patterns differ significantly.	Domain adaptation applied: pre-trained on acted datasets, fine-tuned on TILES acoustic features. Limitation explicitly stated in system documentation.
No real-time data ingestion	System currently operates on batch data. A live hospital deployment would require streaming sensor data ingestion.	FastAPI backend designed for PP2 to handle real-time inference. Current architecture accepts pre-processed feature vectors with no structural changes required for streaming.

5. AI Disclosure

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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- Date: 13.05.2026

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